Pure-Even Multi-k Families and the Arithmetic Core of Resonant Algebra

A constructive contribution to the Inverse Galois Problem

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Abstract

We study Bring–Jerrard quintics $x^5 + \alpha x + \beta$ over $\mathbb{Q}$ whose discriminant is a rational square. Writing $k = \beta/\alpha$, we show that these polynomials are *exactly* the members of the two-parameter family

$$f_{m,k}(x) = x^5 + \alpha(m,k)\,x + k\,\alpha(m,k), \qquad \alpha(m,k) = 256m^2 - \frac{3125k^4}{256}, \quad (m,k) \in \mathbb{Q} \times \mathbb{Q}^\times,$$

with $\operatorname{disc}(f_{m,k}) = (256\,\alpha^2 m)^2$. The family is therefore not a thin special subfamily: $(k, \pm m)$ is a coordinate system on *all* square-discriminant Bring–Jerrard quintics with $\alpha\beta \neq 0$, and pure-evenness holds by construction. A "multi-seed" slice is simply a value of k carrying several seeds. Rescaling $x \mapsto kx$ shows that only $\mu = m/k^2$ matters Galois-theoretically.

We prove an exact discriminant-height formula; give explicit rational paths joining seeds of different ratios; and prove, from Hilbert's irreducibility theorem, that on every slice the generic Galois group over $\mathbb{Q}(m)$ is A_5 (so A_5 fibres have density 1 among integers m, with $O(N^{1/2})$ exceptions up to N). The exceptional irreducible fibres are exactly the D_5 fibres; using an explicit degree-6 resolvent we show they are the rational points of a conic, parametrised by a degree-6 rational function. Reproducible computations on three slices ($k = -8/5,\ -4,\ 4/5$) and a rigid geometric control of opposite parity ($\mathrm{Gal} = S_5$) are reported, with every statement marked as proved, cited, computer-verified, or computational evidence.

The coefficient lattice that motivated this work is the ternary / resonant lattice of an exploratory framework (HQCC / "9 Maths of Unification"). All theorems below are arithmetic and do not depend on any physical interpretation of that lattice. Geometric multi-k fusion (a Nielsen-labelled Hurwitz family recovering several ratios) remains open.

1. Introduction

The Inverse Galois Problem asks which finite groups arise as Galois groups of Galois extensions of $\mathbb{Q}$. Alternating groups A_n are known to occur; the interest of the present note lies in an explicit, organised source of pure-even A_5 polynomials in Bring–Jerrard form and in the arithmetic of that source.

The coefficient source is a ternary numerical lattice that appears in the exploratory resonant-algebraic framework associated with the Hampton qutrit map and the so-called 9 Maths of Unification. That framework is treated here solely as a generator of integers. No claim is made that any physical or cosmological interpretation of the lattice has been established.

Main arithmetic contribution. A transparent pure-even coordinate system (m, k) on the square-discriminant Bring–Jerrard quintics (§3); an exact discriminant-height formula (§3); explicit cross-k paths (§4); a Hilbert-irreducibility theorem showing that A_5 is the generic group on every slice and describing the exceptional D_5 locus explicitly (§5); and reproducible computations (§6).

What the construction is, and is not. Because every square-discriminant Bring–Jerrard quintic lies on exactly one slice (Proposition 3.1), the family does *not* single out an exceptional infinite class of polynomials; its value is organisational. Pure-evenness is built in, the slice parameter $k = \beta/\alpha$ is a convenient normalisation, and the theorems of §5 say what Galois groups to expect along it.

Status of statements. Each result carries one of four tags.

Tag	Meaning
Proved	proved in this note (proof given)
Classical	classical theorem, cited with reference; used as stated
Verified identity	polynomial identity checked by exact arithmetic (sympy / integer linear algebra); no symbolic derivation is given
Evidence	finite computation; not a proof of any statement for all parameters

2. Notation and classical facts

A quintic in Bring–Jerrard form is $f = x^5 + \alpha x + \beta \in \mathbb{Q}[x]$. Its discriminant is

$$\operatorname{disc}(f) = 256\,\alpha^5 + 3125\,\beta^4.$$

Lemma 2.1 (parity). Classical If f is irreducible with $\operatorname{disc}(f) \neq 0$, then $\operatorname{Gal}(f/\mathbb{Q}) \leq A_5$ if and only if $\operatorname{disc}(f)$ is a square in $\mathbb{Q}$.

Lemma 2.2 (even transitive groups; operational A_5 criterion). Proved The transitive subgroups of S_5, up to conjugacy, are C_5, D_5, F_{20}, A_5, S_5. The even ones are exactly C_5, D_5 and A_5, because $F_{20} = \langle (12345), (2354) \rangle$ contains 4-cycles and $F_{20} \cap A_5 = D_5$. Consequently, if f is irreducible, $\operatorname{disc}(f)$ is a square, and for some prime $p \nmid \operatorname{disc}(f)$ the reduction $f \bmod p$ factors with degrees $(3, 1, 1)$, then $\operatorname{Gal}(f/\mathbb{Q}) = A_5$.

Proof. A transitive subgroup of S_5 has order divisible by 5, so it contains a 5-cycle; the groups containing a 5-cycle are C_5, D_5, F_{20}, A_5, S_5 (orders 5, 10, 20, 60, 120). F_{20} is generated by a 5-cycle and a 4-cycle, hence is not even; its subgroup $F_{20} \cap A_5$ has index 2, order 10, so equals D_5 (it is generated by the 5-cycle (12345) and $(2354)^2 = (25)(34)$). By Dedekind's theorem the degrees of the irreducible factors of $f \bmod p$, for $p \nmid \operatorname{disc}(f)$, are the cycle lengths of a Frobenius element of $G = \operatorname{Gal}(f/\mathbb{Q})$. A Frobenius element of type $(3, 1, 1)$ is a 3-cycle, of order 3. Since $G \leq A_5$ is transitive and $3 \nmid 5, 10$, $G = A_5$. ∎

(Chebotarev's theorem gives the converse: for $G = A_5$, 3-cycles occur with density $20/60 = 1/3$ among primes, whereas C_5 and D_5 never contain one.)

Remark (no C_5). For an irreducible Bring–Jerrard quintic $\operatorname{Gal} \neq C_5$: a cyclic group of order 5 has no element of order 2, so complex conjugation would be trivial and all five roots real; but $f' = 5x^4 + \alpha$ has at most two real zeros, so f has at most three real roots. Hence an irreducible Bring–Jerrard quintic with square discriminant has $\operatorname{Gal} \in \{D_5, A_5\}$.

Lemma 2.3 (homogenisation / rescaling). Proved For $t \in \mathbb{Q}^\times$ put $f_t(x) = x^5 + \alpha t^4 x + \beta t^5 = t^5 f(x/t)$. Then $\operatorname{disc}(f_t) = t^{20} \operatorname{disc}(f)$ (weight 20 of the discriminant under $(\alpha, \beta) \mapsto (\alpha t^4, \beta t^5)$), f_t and f have the same splitting field, and f_t is pure-even whenever f is. The ratio changes as $k \mapsto tk$.

3. The pure-even multi-k theorem

Proposition 3.1 (the family is a coordinate system on all square-discriminant quintics). Proved Let $f = x^5 + \alpha x + \beta \in \mathbb{Q}[x]$ with $\operatorname{disc}(f) \neq 0$.

1. If $\alpha = 0$ then $\operatorname{disc}(f) = 3125\beta^4 = 5\,(25\beta^2)^2$ is never a rational square. Hence a square discriminant forces $\alpha \neq 0$.
2. If $\alpha \neq 0$ and $k = \beta/\alpha$, then $\operatorname{disc}(f) = \alpha^4\,(256\alpha + 3125k^4)$, so $\operatorname{disc}(f)$ is a square if and only if $256\alpha + 3125k^4$ is a rational square.
3. In that case there is a unique $m \in \mathbb{Q}$ up to sign with $256\alpha + 3125k^4 = (256m)^2$, namely $m = \sqrt{256\alpha + 3125k^4}\,/\,256$; then $\alpha = \alpha(m,k) = 256m^2 - 3125k^4/256$, $\beta = k\alpha$, i.e. $f = f_{m,k}$, and $m \neq 0$ because $\operatorname{disc} \neq 0$.
4. Conversely, for every $(m,k) \in \mathbb{Q} \times \mathbb{Q}^\times$ one has $\alpha(m,k) \neq 0$, and $f_{m,k}$ has square discriminant $(256\alpha^2 m)^2$.

Hence $(m,k) \mapsto f_{m,k}$ is a bijection from $(\mathbb{Q}^\times \times \mathbb{Q})/\{m \sim -m\}$ onto the set of Bring–Jerrard quintics with nonzero square discriminant. The slice $k = 0$ consists of reducible polynomials $x(x^4 + 256m^2)$, so irreducible members have $k \neq 0$.

Proof. (1) is immediate since 5 is not a rational square. (2): substitute $\beta = k\alpha$ into $256\alpha^5 + 3125\beta^4$. (3): a square $256\alpha + 3125k^4 = s^2$ gives $m = s/256$, and solving for α gives the stated formula; uniqueness up to sign is clear. (4): $\alpha(m,k) = 0$ would mean $65536\,m^2 = 3125\,k^4$, i.e. $5 = \big(256m/(25k^2)\big)^2$, impossible in $\mathbb{Q}$ for $k \neq 0$. Finally $256\alpha + 3125k^4 = 65536m^2$, so $\operatorname{disc} = \alpha^4 \cdot 65536m^2 = (256\alpha^2 m)^2$. ∎

In particular, "multi-seed" has a purely elementary meaning: a *seed* is a member $f_{m,k}$ that is irreducible with Galois group A_5, and a slice is *multi-seed* if it carries two or more seeds. Section 5 shows that every slice $k \neq 0$ carries infinitely many.

Theorem 3.2 (pure-even k-slice). Proved Fix $k \in \mathbb{Q}^\times$. For every $m \in \mathbb{Q}$,

$$\operatorname{disc}\big(x^5 + \alpha(m)x + \beta(m)\big) = (256\,\alpha(m)^2\,m)^2, \qquad \alpha(m) = 256m^2 - \tfrac{3125k^4}{256},\ \ \beta(m) = k\,\alpha(m).$$

In particular every irreducible specialisation has even Galois group, at most A_5.

Proof. Substituting $\beta = k\alpha$ gives $\operatorname{disc} = 256\alpha^5 + 3125(k\alpha)^4 = \alpha^4(256\alpha + 3125k^4)$. With $\alpha = 256m^2 - 3125k^4/256$,

$$256\alpha + 3125k^4 = 256^2 m^2 - 3125k^4 + 3125k^4 = 65536\,m^2,$$

so $\operatorname{disc} = \alpha^4 \cdot 65536\,m^2 = (256\alpha^2 m)^2$. ∎

Corollary 3.3. Proved The identity of Theorem 3.2 is a polynomial identity over $\mathbb{Z}[1/2]$, hence holds over every field of characteristic $\neq 2$ (the factor 256 must be invertible to define m).

Proposition 3.4 (normal form; only $\mu = m/k^2$ matters). Proved For $t \in \mathbb{Q}^\times$, $t^5 f_{m,k}(x/t) = f_{t^2 m,\,tk}(x)$. Consequently, with $\mu = m/k^2$,

$$f_{m,k}(x) = k^5\, g_\mu(x/k), \qquad g_\mu(y) = y^5 + \Big(256\mu^2 - \tfrac{3125}{256}\Big)(y+1),$$

so $f_{m,k}$ and $g_{m/k^2} = f_{m/k^2,1}$ have the same splitting field. Moreover $-f_{m,k}(-x) = f_{m,-k}(x)$, so k and $-k$ give isomorphic fields for the same m.

Proof. Both sides are polynomials in x; compare coefficients using $\alpha(t^2 m, tk) = t^4\alpha(m,k)$ (since $256t^4m^2 - 3125t^4k^4/256$) and $tk\,\alpha(t^2m, tk) = t^5 k\alpha(m,k)$. Take $t = k$ for the normal form. For $x \mapsto -x$: $-f_{m,k}(-x) = x^5 + \alpha x - k\alpha$ and $\alpha(m,-k) = \alpha(m,k)$. ∎

So the ratio k is a normalisation of scale: the Galois-theoretic content of all slices is carried by the single line $\mu \in \mathbb{Q}^\times/\pm$, and a slice k is that line viewed through the rescaling $m = k^2\mu$.

Theorem 3.5 (discriminant height, precise form). Proved Fix $k \in \mathbb{Q}^\times$ and put $\varepsilon = \dfrac{3125\,k^4}{65536\,m^2}$. For all $m \neq 0$,

$$\operatorname{disc}(f_{m,k}) = 2^{48}\, m^{10}\,(1-\varepsilon)^4.$$

Hence for $|m| > 25\sqrt{5}\,k^2/256$ (equivalently $\varepsilon < 1$),

$$\log|\operatorname{disc}(f_{m,k})| = 10\log|m| + 48\log 2 + 4\log(1-\varepsilon) = 10\log|m| + 48\log 2 - \frac{3125\,k^4}{16384}\,m^{-2} + O(m^{-4}),$$

and for $\varepsilon \le \frac{1}{2}$ the remainder satisfies $0 \le \big(-4\log(1-\varepsilon) - 4\varepsilon\big) \le 4\varepsilon^2$. In particular $\log|\operatorname{disc}(f_m)| = 10\log|m| + 48\log 2 + O(m^{-2})$ along a fixed slice. (Here disc is the discriminant of the monic polynomial $f_{m,k} \in \mathbb{Q}[x]$; for the integer polynomial obtained by clearing denominators with Lemma 2.3 it is multiplied by t^{20}.)

Proof. Write $\alpha = 256m^2(1-\varepsilon)$, since $\dfrac{3125k^4/256}{256m^2} = \varepsilon$. By Theorem 3.2, $\operatorname{disc} = 65536\,\alpha^4 m^2 = 2^{16}\cdot 2^{32} m^8 (1-\varepsilon)^4 \cdot m^2 = 2^{48} m^{10}(1-\varepsilon)^4$. Taking $\log|\cdot|$ and expanding $-\log(1-\varepsilon) = \sum_{j\ge 1}\varepsilon^j/j$ gives the rest; for $\varepsilon \le \frac{1}{2}$, $\sum_{j\ge 2}\varepsilon^j/j \le \frac{\varepsilon^2}{2(1-\varepsilon)} \le \varepsilon^2$. And $4\varepsilon = \dfrac{3125k^4}{16384\,m^2}$. ∎

4. Seeds, multi-seed slices and cross-k paths

Seeds. The four polynomials of Appendix A are the principal seeds used in this note. They sit on the slices flagship $k = -\frac{8}{5}$, LSW $k = -4$, classical $k = \frac{4}{5}$.

polynomial	k	m	$\mu = m/k^2$	Gal
$x^5 - 55x + 88$ (flagship)	$-8/5$	$5/16$	$125/1024$	A_5
$x^5 - 100x + 400$ (LSW)	-4	$55/16$	$55/256$	A_5
$x^5 + 20x + 16$ (classical)	$4/5$	$5/16$	$125/256$	A_5
$x^5 + 20x - 32$	$-8/5$	$5/8$	$125/512$	D_5

(Appendix A gives discriminants and Frobenius data.)

A bounded integer search. Evidence To exhibit multi-seed slices without any hypothesis on the origin of the coefficients, we enumerated all integer pairs (α, β) with $1 \le |\alpha| \le 1000$, $1 \le \beta \le 5000$ (the mirror $\beta \mapsto -\beta$ is $x \mapsto -x$ and sends $k \mapsto -k$) such that $256\alpha^5 + 3125\beta^4$ is a nonzero square. There are 55 such pairs; all are irreducible; 48 have $\mathrm{Gal} = A_5$ and 7 have $\mathrm{Gal} = D_5$ (Frobenius cycle types, primes $7 \le p \le 4000$, cross-checked with PARI `polgalois`; no disagreement). The 48 seeds fall on 18 distinct ratios $k = \beta/\alpha$, of which 7 carry at least two seeds:

k	seeds (α, β) in the search box
-4	$(-100, 400), (-209, 836), (-316, 1264), (-421, 1684), (-524, 2096), (-625, 2500), (-724, 2896), (-821, 3284), (-916, 3664)$
4	$(124, 496), (239, 956), (356, 1424), (475, 1900), (596, 2384), (719, 2876), (844, 3376), (971, 3884)$
$4/5$	$(20, 16), (95, 76), (220, 176), (395, 316), (620, 496), (895, 716)$
$8/5$	$(145, 232), (320, 512), (545, 872), (820, 1312)$
$-16/5$	$(-55, 176), (-380, 1216), (-655, 2096), (-880, 2816)$
$-12/5$	$(-180, 432), (-305, 732), (-380, 912)$
$12/5$	$(220, 528), (495, 1188), (820, 1968)$

The flagship $x^5 - 55x + 88$ is the only seed with $k = -8/5$ in this box, so within the box its slice is not multi-seed; the same holds for the other ratios not listed. These counts depend on the box. By Theorem 5.1, *every* slice contains infinitely many seeds, so "multi-seed" is not an intrinsic property of k.

Two-parameter envelope. By Proposition 3.1 the pure-even envelope $\alpha(m, s) = 256m^2 - 3125s^4/256$, $\beta = s\alpha$, $\operatorname{disc} = (256\alpha^2 m)^2$ is the whole set of square-discriminant quintics; freezing $s = k$ recovers a slice.

Proposition 4.1 (explicit cross-k paths). Proved (direct substitution; also checked by exact arithmetic, Verified identity.) Let $u \in \mathbb{Q}$ and $\alpha(u) = \alpha(m(u), k(u))$. Each of the following is a path of pure-even quintics $x^5 + \alpha(u)x + k(u)\alpha(u)$, with $\mathrm{disc} = (256\alpha(u)^2 m(u))^2$ for every u, joining two seeds.

path	$k(u)$	$m(u)$	$\alpha(u)$	endpoints
flagship → classical	$-\frac{8}{5} + \frac{12}{5}u$	$\frac{5}{16}$ (constant)	$-5\,(81u^4 - 216u^3 + 216u^2 - 96u + 11)$	$u=0$: $\alpha=-55, \beta=88$; $u=1$: $\alpha=20, \beta=16$
flagship → LSW	$-\frac{8}{5} - \frac{12}{5}u$	$\frac{5+50u}{16}$	$-5\,(81u^4 + 216u^3 - 284u^2 - 4u + 11)$	$u=0$: $(-55, 88)$; $u=1$: $\alpha=-100, \beta=400$
LSW → classical	$-4 + \frac{24}{5}u$	$\frac{55-50u}{16}$	$-20\,(324u^4 - 1080u^3 + 1225u^2 - 475u + 5)$	$u=0$: $(-100, 400)$; $u=1$: $(20, 16)$

Proof. The disc identity is Theorem 3.2 evaluated at $(m(u), k(u))$. For the first path, $\alpha(u) = 256 \cdot \frac{25}{256} - \frac{3125}{256}k(u)^4 = 25 - \frac{3125}{256}k(u)^4$; at $k = -\frac{8}{5}$, $k^4 = \frac{4096}{625}$ so $\alpha = 25 - 80 = -55$, $\beta = -\frac{8}{5}(-55) = 88$; at $k = \frac{4}{5}$, $k^4 = \frac{256}{625}$ so $\alpha = 25 - 5 = 20$, $\beta = \frac{4}{5} \cdot 20 = 16$. For the others, $m = \frac{55}{16}$ at LSW: $\alpha(\frac{55}{16}, -4) = 3025 - 3125 = -100$, $\beta = 400$. ■

A path with constant m cannot join flagship ($m = 5/16$) to LSW ($m = 55/16$), which is why $m(u)$ varies in the second and third paths; the first path is the exceptional case in which both seeds have the same $m = 5/16$.

Remark (structural). Any path $u \mapsto (m(u), k(u))$ with $m, k \in \mathbb{Q}(u)$ is pure-even, since pure-evenness holds at *every* point of the (m, k)-plane. The paths above carry no further Galois-theoretic information by themselves; they illustrate that seeds of different ratios lie on a common rational curve of square-discriminant quintics. By Corollary 5.2 below, along each such path with one A_5 specialisation the generic group is again A_5.

5. Galois groups along a slice

5.1 Specialisation and Hilbert irreducibility

Proposition 5.0 (specialisation). Classical Let $F(m, x) \in \mathbb{Q}[m][x]$ be monic in x with $\mathrm{disc}_x F \neq 0$, let $G = \mathrm{Gal}(F/\mathbb{Q}(m))$ acting on the roots, and let $m_0 \in \mathbb{Q}$ with $\mathrm{disc}_x F(m_0) \neq 0$ (so the coefficients have no pole at m_0 and $F(m_0, x)$ is separable). Then $\mathrm{Gal}(F(m_0, x)/\mathbb{Q})$ embeds in G as a permutation group, uniquely up to conjugacy in G; and if $F(m_0, x)$ is irreducible then F is irreducible over $\mathbb{Q}(m)$. (See Serre, *Topics in Galois Theory*, §3.3; Lang, *Fundamentals of Diophantine Geometry*, Ch. 9; Zywina, arXiv:1011.6465, §1.1.)

(The embedding is the inclusion of a decomposition group at the unramified prime $m - m_0$. Irreducibility lifts because a monic factorisation over $\mathbb{Q}(m)$ lies in $\mathbb{Q}[m][x]$ by Gauss's lemma and specialises to a factorisation of the same degrees.)

For $f_{m,k}$ the coefficients are polynomials in m, and $\mathrm{disc}_x f_{m,k} = (256\alpha^2 m)^2$ vanishes only at $m = 0$ (Proposition 3.1(4)). So Proposition 5.0 applies at every $m_0 \in \mathbb{Q}^\times$.

Theorem 5.1 (generic group A_5; Hilbert irreducibility consequence). Proved from Classical inputs. Let $k \in \mathbb{Q}^\times$ and suppose that for some $m_0 \in \mathbb{Q}^\times$ the polynomial $f_{m_0,k}$ is irreducible with $\mathrm{Gal} = A_5$ (this holds, for instance, for $k = -\frac{8}{5}, -4, \frac{4}{5}$ at the seeds of §4). Then:

1. $\mathrm{Gal}(f_{m,k}/\mathbb{Q}(m)) = A_5$.
2. The set $\Omega_k = \{m \in \mathbb{Q}^\times : \ f_{m,k} \text{ is reducible or } \mathrm{Gal}(f_{m,k}/\mathbb{Q}) \neq A_5\}$ is a thin subset of $\mathbb{Q}$ in the sense of Serre.
3. $\#\{m \in \mathbb{Z} : \ 1 \le |m| \le N, \ m \in \Omega_k\} = O_k\big(N^{1/2}\big)$. In particular the fibres with $\mathrm{Gal} = A_5$ have density 1 among the integers m.
4. By Proposition 3.4 the hypothesis holds for **every** $k \in \mathbb{Q}^\times$ as soon as it holds for one: $\mathrm{Gal}(f_{m,k}/\mathbb{Q}(m)) = A_5$ for all $k \neq 0$.

Proof. (1) $f = f_{m,k}$ is irreducible over $\mathbb{Q}(m)$ by Proposition 5.0. Its discriminant $(256\alpha^2 m)^2$ is a square in $\mathbb{Q}(m)$, so $G = \mathrm{Gal}(f/\mathbb{Q}(m)) \le A_5$ (Lemma 2.1 over the field $\mathbb{Q}(m)$). By Proposition 5.0 (with $\mathrm{disc}(f_{m_0,k}) \neq 0$) G contains a conjugate of $\mathrm{Gal}(f_{m_0,k}/\mathbb{Q}) = A_5$. Hence $G = A_5$. (2) For $G = \mathrm{Gal}(f/\mathbb{Q}(m))$, the set of $m_0 \in \mathbb{Q}$ outside the finite pole/non-separability set at which the specialised group is a proper subgroup of G is thin (Serre, *Topics*, Prop. 3.3.5; Hilbert's irreducibility theorem), and the set where f_{m_0} is reducible is thin for the same reason. Adding the point $m = 0$ preserves thinness. (3) For a thin subset of $\mathbb{Q}$ the number of integers of absolute value $\le N$ is $O(N^{1/2})$ (Serre, *Topics*, Thm. 3.4.4; *Lectures on the Mordell–Weil theorem*, §9.7, a consequence of Siegel's theorem; the n-variable large-sieve version with a log factor is S. D. Cohen, Proc. London Math. Soc. (3) 43 (1981), 227–250). (4) Proposition 3.4 gives $\mathrm{Gal}(f_{m,k}/\mathbb{Q}(m)) = \mathrm{Gal}(f_{m/k^2,1}/\mathbb{Q}(m))$ and $m \mapsto m/k^2$ is an automorphism of $\mathbb{Q}(m)$ over $\mathbb{Q}$; so the generic group is independent of k. ∎

This settles, on the three slices of §4 (and, by (4), on every slice), the density of irreducible A_5 fibres among integer m: it is 1. It does not give a bound on the *first* exceptional m, and the implied constant is not effective as stated.

Corollary 5.2 (paths). Proved Let $u \mapsto (m(u), k(u))$ be a path with $m, k \in \mathbb{Q}(u)$, $k \neq 0$, and suppose that at some u_0 (no pole, $m(u_0) \neq 0$) the fibre $f_{m(u_0),k(u_0)}$ is irreducible with $\mathrm{Gal} = A_5$. Then the generic group of the path over $\mathbb{Q}(u)$ is A_5 and the same thin-set conclusions hold in the variable u. This applies to the three paths of Proposition 4.1.

5.2 The D_5 locus

Theorem 5.3 (degree-6 resolvent criterion). Classical (Dummit, *Solving solvable quintics*, Math. Comp. 57 (1991) 387–401; Spearman–Williams, Amer. Math. Monthly 101 (1994) 986–992) Let $f \in \mathbb{Q}[x]$ be an irreducible quintic and R_f its sextic resolvent, the polynomial whose roots are $\theta_j = \left(\sum_i x_i x_{i+1} - \sum_i x_i x_{i+2}\right)^2$ (indices mod 5, evaluated on the six F_{20}-cosets of orderings of the roots x_i). If R_f is squarefree, then $\mathrm{Gal}(f/\mathbb{Q})$ is contained in a conjugate of F_{20} if and only if R_f has a rational root. Consequently, if moreover $\mathrm{disc}(f)$ is a square, then $\mathrm{Gal}(f/\mathbb{Q}) \le D_5$ if and only if R_f has a rational root.

Why: S_5 acts on the six cosets of F_{20} (as $PGL_2(\mathbb{F}_5)$ on the projective line); the stabiliser of a coset is a conjugate of F_{20}. If the six θ_j are distinct, a rational θ_j is fixed by G, hence G fixes the coset j; conversely, a group inside a conjugate of F_{20} fixes the corresponding θ_j. For even G use $F_{20} \cap A_5 = D_5$ (Lemma 2.2).

For $f = x^5 + ax + b$ the resolvent can be written down.

Proposition 5.4 (explicit resolvent). Verified identity For $f = x^5 + ax + b$,

$$R(X; a, b) = X^6 - 40a\,X^5 + 880a^2X^4 - 8960a^3X^3 + 44800a^4X^2 - 1024\,(106a^5 + 3125b^4)\,X + 102400\,a^6,$$

and $\mathrm{disc}_X\, R = 2^{60}\,5^{20}\,b^{12}\;\mathrm{disc}(f)^3$. In particular R is squarefree if and only if $b \neq 0$ and $\mathrm{disc}(f) \neq 0$.

How it was verified. The coefficients of R lie in $\mathbb{Z}[a, b]$, are weighted homogeneous ($\mathrm{wt}\, a = 4$, $\mathrm{wt}\, b = 5$, $\mathrm{wt}\, X = 4$), and so involve few monomials. We computed the six values θ_j at 60 random integer points (a, b) to 300 digits, rounded the coefficients of $\prod(X - \theta_j)$ to integers (all within 10^{-100} of integers), solved the overdetermined linear system for the integer coefficients exactly, and re-checked the resulting formula at the remaining points and at 10 fresh points of size up to 200. The identity is therefore established by exact interpolation of rigorously rounded values; no symbolic derivation is given here (script `scripts/resolvent.py`). The criterion of Theorem 5.3 is used only in this explicit form.

Checks. The resolvent has no rational root for the three A_5 seeds of Appendix A, and has the rational root $X = 400$ for $x^5 + 20x - 32$ (which is $X = \alpha Y$ with $Y = 20$, the value $Y(\lambda = 0)$ in Theorem 5.5) (a D_5 fibre). In the scan of §6 the criterion agrees with PARI's `polgalois` on all 1830 polynomials tested.

On the slice $b = k\alpha$, put $X = \alpha Y$ and $s = 65536m^2/\alpha$. Then $R(\alpha Y) = \alpha^6\big(P_0(Y) - 1024\,s\,Y\big)$ with

$$P_0(Y) = G(Y)^2, \quad G(Y) = Y^3 - 20Y^2 + 240Y + 320, \qquad P_0(Y) - 2^{18}Y = (Y - 4)^4(Y^2 - 24Y + 400).$$

Theorem 5.5 (the D_5 fibres are the rational points of a conic). **Proved** from Theorem 5.3 and Proposition 5.4. Let $k \in \mathbb{Q}^\times$, $m \in \mathbb{Q}^\times$, and suppose $f_{m,k}$ is irreducible. Then $\mathrm{Gal}(f_{m,k}/\mathbb{Q}) \neq A_5$ (that is, $= D_5$, by the remark after Lemma 2.2) if and only if there is $\lambda \in \mathbb{Q} \setminus \{-1\}$ with

$$m = \pm\, k^2 M(\lambda), \qquad M(\lambda) = -\frac{(\lambda^3 - 15\lambda^2 - 25\lambda - 25)(\lambda^3 + 5\lambda^2 + 25\lambda + 25)}{512\,(\lambda+1)^2\,(\lambda^2 + 2\lambda + 5)}.$$

Equivalently: $\mathrm{Gal} = D_5$ iff there is a rational point (Y, w) with $Y \notin \{0, 4\}$ on the conic $w^2 = 5(Y^2 - 24Y + 400)$ and $m = \pm 125\,k^2\,G(Y)/\big(256\,w\,(Y-4)^2\big)$; the point (Y, w) is $\big(4 - \frac{80(\lambda+1)}{\lambda^2-5},\ 40 + \lambda(Y-4)\big)$. Moreover the numerator cubics have no rational roots, so $M(\lambda) \neq 0$ for $\lambda \in \mathbb{Q} \setminus \{-1\}$.

Proof. By Theorem 5.3 and Proposition 5.4 (R is squarefree since $b = k\alpha \neq 0$ and $\mathrm{disc} \neq 0$), and since $\mathrm{disc}(f_{m,k})$ is a square, the group is $\leq D_5$ iff R has a rational root $X = \alpha Y$. Since $P_0(0) = 102400 \neq 0$ we have $Y \neq 0$, so $s = G(Y)^2/(1024\,Y)$ (note $G(Y) \neq 0$ for rational Y, as the cubic has no rational root). From $\alpha = 256m^2 - c$, $c = 3125k^4/256$, and $\alpha = 65536m^2/s$ we get $m^2 = \dfrac{c\,s}{256\,(s - 256)}$, and $s - 256 = (Y-4)^4(Y^2 - 24Y + 400)/(1024\,Y)$, so

$$m^2 = \frac{3125\,k^4\,G(Y)^2}{65536\,(Y-4)^4\,(Y^2 - 24Y + 400)}.$$

Because $3125 = 5 \cdot 25^2$, m is rational iff $5(Y^2 - 24Y + 400) = w^2$ for a rational w, and then $m = \pm 125k^2 G(Y)/(256\,w\,(Y-4)^2)$. Conversely, every such (Y, w) with $Y \neq 0, 4$ defines m and $s = G^2/(1024Y) \neq 0, 256$ for which the displayed relations hold in reverse, so $R(\alpha Y) = 0$. The conic has the rational point $(4, 40)$; projecting from it, the line $w = 40 + \lambda(Y-4)$ meets the conic again at $Y - 4 = -80(\lambda+1)/(\lambda^2 - 5)$. Every rational point with $Y \neq 4$ arises from exactly one $\lambda \in \mathbb{Q} \setminus \{-1\}$ (the tangent at $(4, 40)$ is $\lambda = -1$; $Y = 0$ would need $\lambda^2 - 20\lambda - 25 = 0$, which has no rational root). Substituting gives the formula for M. ∎

Examples (**Evidence** for the match, checked by exact arithmetic): $x^5 + 20x - 32$ ($k = -\frac{8}{5}$, $m = \frac{5}{8}$) corresponds to $\lambda = 0$ (and -5), since $M(0) = \frac{125}{512}$ and $k^2 M(0) = \frac{64}{25} \cdot \frac{125}{512} = \frac{5}{8}$; the D_5 fibres found in §6 are $(k, m) = (-\frac{8}{5}, \frac{5}{8})$ [$\lambda = 0, -5$], $(-\frac{8}{5}, \frac{5}{9})$ [$\lambda = 5, -\frac{5}{3}$], $(-4, \frac{7}{2})$ [$\lambda = 1, -3$].

Consequences. (i) Every slice $k \neq 0$ contains infinitely many D_5 fibres: $m = 125k^2/512$ ($\lambda = 0$) is a rescaling of $x^5 + 20x - 32$, so the family over $\mathbb{Q}(\lambda)$ is irreducible and all but a thin set of λ give irreducible fibres; they all lie in the thin exceptional set of Theorem 5.1, being the image of $\mathbb{P}^1(\mathbb{Q})$ under the degree-6 map $\lambda \mapsto k^2 M(\lambda)$. (ii) On an irreducible even fibre, $\mathrm{Gal} = A_5$ if and only if the explicit sextic of Proposition 5.4 has no rational root. (iii) Hence A_5 is certified on the scanned fibres below by either a Frobenius 3-cycle (Lemma 2.2) or the absence of a rational root of R.

6. Computational evidence

Method. **Evidence** For each slice $k \in \{-\frac{8}{5}, -4, \frac{4}{5}\}$ and each test value m, the rational polynomial $f_{m,k}$ was replaced (Lemma 2.3, minimal t) by an integer polynomial $x^5 + Ax + B$ with the same Galois group; its irreducibility was tested over $\mathbb{Q}$ (FLINT); its Frobenius cycle types were computed for all primes $7 \leq p \leq 4000$ not dividing disc (542–547 primes used per polynomial); and its group was classified as

- A_5 if irreducible and a type $(3, 1, 1)$ occurred (certified by Lemma 2.2, since all discriminants are squares);
- D_5 if irreducible, only the types (1^5), (5), $(2, 2, 1)$ occurred, and $(2, 2, 1)$ occurred;
- **undetermined** otherwise.

Every classification was independently cross-checked against PARI/GP `polgalois`, and, for irreducible polynomials, against the resolvent criterion of Theorem 5.3 with the explicit R of Proposition 5.4. Test sets: **(I)** all integers $1 \leq m \leq 300$ (the polynomial for $-m$ is identical, so $|m| \leq 300$ means 300 distinct polynomials per slice, 600 values of m); **(R)** all reduced fractions $m = p/q$ with $2 \leq q \leq 10$ and $0 < m \leq 10$ (310 per slice). Condition $\alpha \neq 0$ holds automatically (Proposition 3.1).

Table 1. Galois groups of the tested fibres.

slice k	test set	tested	irreducible	A_5	D_5	undetermined	reducible
$-8/5$	(I) integers 1..300	300	300	300	0	0	0
-4	(I)	300	300	300	0	0	0
$4/5$	(I)	300	300	300	0	0	0
$-8/5$	(R) rationals	310	310	308	2	0	0
-4	(R)	310	310	309	1	0	0
$4/5$	(R)	310	310	310	0	0	0
total		1830	1830	1827	3	0	0

There were no disagreements between the Frobenius classification, PARI's `polgalois`, and the resolvent criterion on any of the 1830 polynomials. The three D_5 fibres are $(k,m) = (-\frac{8}{5}, \frac{5}{8})$, $(-\frac{8}{5}, \frac{5}{9})$, $(-4, \frac{7}{2})$, i.e. $x^5 + 20x - 32$, $x^5 - 80x + 384$, $x^5 + 11x - 44$, and they are exactly the points of the scanned set picked out by the parametrisation of Theorem 5.5 (searched over $\lambda = p/q$, $|p|, q \le 600$).

Extended integer range. Using Theorem 5.3 and Proposition 5.4 (irreducible, no rational root of $R \Rightarrow A_5$): for each of the three slices all integers $1 \le m \le 5000$ give an irreducible polynomial with Galois group A_5 (5000 of 5000; no reducible and no D_5 fibre). A search over $\lambda = p/q$ with $|p|, q \le 1200$ found no integer m of the form $\pm k^2 M(\lambda)$ on these three slices, so no integer D_5 fibre was found. This is consistent with Theorem 5.1(3), but is not a proof that none exists.

Negative control. The rigid Belyi cover $\varphi(y) = 6y^5 - 15y^4 + 10y^3$ has $\varphi = y^3(6y^2 - 15y + 10)$, $\varphi - 1 = (y-1)^3(6y^2 + 3y + 1)$, a 5-cycle at infinity (ramification $(3,1,1), (3,1,1), (5)$) and geometric monodromy A_5. For the monic polynomial $\mathrm{monic}(\varphi(y) - t) = y^5 - \frac{5}{2}y^4 + \frac{5}{3}y^3 - \frac{t}{6}$ one has, exactly (Verified identity),

$$\mathrm{disc}\big(\mathrm{monic}(\varphi(y) - t)\big) = 5\Big(\frac{25\,t(t-1)}{36}\Big)^2,$$

so the discriminant is $5\times$ a square for rational $t \notin \{0, 1\}$. The fibre at $t = 3$ (integer model $z^5 - 15z^4 + 60z^3 - 3888$, $z = 6y$) is irreducible with discriminant $5 \cdot 251942400^2$ and Frobenius cycle types $(5), (4,1), (3,2), (3,1,1), (2,2,1), (2,1,1,1), (1^5)$ all occurring over $7 \le p \le 4000$ (including 4-cycles and a transposition type), so $\mathrm{Gal} = S_5$; PARI `polgalois` agrees (each cycle type that occurs is realised by an element of the group, by Dedekind; a transitive subgroup of S_5 containing a 3-cycle and an odd permutation is S_5). Over the integers $t = 2, \dots, 100$, PARI gives S_5 for 98 fibres and one reducible fibre ($t = 32$). This confirms that the detection pipeline correctly reports odd monodromy when evenness fails, and that the pure-even construction is what delivers A_5 in §§3–5.

Other statements. The discriminant identities (Theorems 3.2, 3.5; Propositions 3.4, 4.1) were additionally confirmed by exact symbolic computation, and the height expansion numerically (remainder $/\varepsilon^2 \to -2$).

7. What this paper does not claim

1. A general solution of the Inverse Galois Problem.
2. Priority for the existence of A_5 or A_6 extensions of $\mathbb{Q}$ (both are classical).
3. A structural theorem that the ternary lattice must produce alternating monodromy.
4. Geometric multi-k fusion: no Nielsen-labelled Hurwitz family has yet been shown to recover several distinct ratios as Bring–Jerrard fibres.
5. Any physical, cosmological, or cryptographic conclusion drawn from the resonant lattice or from the broader "9 Maths of Unification" framework. Those interpretations lie outside the scope of the theorems above.
6. That the pure-even family is a thin or special subfamily of Bring–Jerrard quintics: by Proposition 3.1 it is all of the square-discriminant locus, and by Proposition 3.4 the ratio k is a normalisation of scale.
7. Anything about the contents of any external repository; Appendix B lists exactly what accompanies this note.

8. Open problems

1. Integral exceptions. Determine whether there is an integer m with $\mathrm{Gal}(f_{m,k}) \neq A_5$ on the slices $k = -\frac{8}{5}, -4, \frac{4}{5}$; by Theorem 5.5 this is the Diophantine question whether $\pm k^2 M(\lambda) \in \mathbb{Z}$ for some $\lambda \in \mathbb{Q}$. (None was found for $\lambda = p/q$, $|p|, q \leq 1200$.) Make the $O(N^{1/2})$ bound effective.
2. Quantitative form of the D_5 locus: count the D_5 fibres of bounded height on a slice via the degree-6 map $\lambda \mapsto k^2 M(\lambda)$ of Theorem 5.5, and obtain a symbolic derivation of the resolvent of Proposition 5.4 (currently a verified identity).
3. Construct a geometric (Nielsen-labelled) family whose Bring–Jerrard ratios include at least two catalogue seeds on distinct slices, i.e. a family not obtained from the coordinate system of Proposition 3.1.
4. Extract axioms on ternary matrix templates that force square discriminant (currently a search criterion, not a theorem for the full template class).

9. Conclusion

The pure-even multi-k theory supplies an explicit, organised and computationally checked source of even Galois representations of degree 5 over $\mathbb{Q}$. It is a coordinate system on all square-discriminant Bring–Jerrard quintics rather than a special subfamily; within it, A_5 is the generic group on every slice, the exceptional irreducible fibres are the D_5 fibres, and these form an explicit conic-parametrised set. The ternary lattice that generates the coefficients is of independent exploratory interest; the theorems themselves are ordinary statements in constructive algebraic number theory and stand without reference to any physical model. Geometric fusion and structural necessity remain open. They are natural directions for further work; they are not required for the validity or citability of the results proved here.

Appendix A. Principal seeds

Each polynomial was checked to be irreducible (FLINT and PARI), to have square discriminant, and classified by Frobenius cycle types over the primes $7 \leq p \leq 4000$ not dividing the discriminant (546–547 primes), with PARI `polgalois` agreeing.

polynomial	k, m	disc	**Frobenius types** (5) / $(3,1,1)$ / $(2,2,1)$ / (1^5)	Gal
$x^5 - 55x + 88$	$-\frac{8}{5},\ \frac{5}{16}$	$242000^2 = 2^8 5^6 11^4$	210 / 206 / 118 / 12	A_5
$x^5 - 100x + 400$	$-4,\ \frac{55}{16}$	$8800000^2 = 2^{16} 5^{10} 11^2$	221 / 180 / 138 / 7	A_5
$x^5 + 20x + 16$	$\frac{4}{5},\ \frac{5}{16}$	$32000^2 = 2^{16} 5^6$	221 / 181 / 138 / 7	A_5
$x^5 + 20x - 32$	$-\frac{8}{5},\ \frac{5}{8}$	$64000^2 = 2^{18} 5^6$	221 / 0 / 279 / 47	D_5

For $x^5 + 20x - 32$ the absence of $(3,1,1)$ is evidence only; $\mathrm{Gal} \leq D_5$ is proved by the rational root $X = 400$ of the resolvent (Theorem 5.3, Proposition 5.4), and since $\mathrm{Gal} \neq C_5$ (remark after Lemma 2.2), $\mathrm{Gal} = D_5$. For the three A_5 seeds the 3-cycle type proves $\mathrm{Gal} = A_5$ (Lemma 2.2) and the resolvent has no rational root.

Appendix B. Reproducibility

All numbers in §§3–6 were produced by the Python scripts distributed with this note in the folder `scripts/` (outputs in `data/`), run with `OMP_NUM_THREADS=1` and computing only with exact integer/rational arithmetic, apart from the 300-digit interpolation of Proposition 5.4 and the numerical check of the height expansion. Software: Python 3.13, SymPy 1.14, python-flint 0.9, cypari2 2.2.4 (PARI/GP), mpmath 1.3. Script `run_all.sh` regenerates every table; the whole run takes under a minute on four cores. This note makes no statement about the contents of any other repository.

script	produces
`verify_algebra.py`	Props. 3.1, 3.4, 4.1, Thms. 3.2, 3.5 (exact symbolic checks)
`resolvent.py`	Prop. 5.4 (sextic resolvent), `data/sextic_resolvent.txt`
`d5_locus.py`, `d5_integer_search.py`, `d5_crosscheck.py`	Thm. 5.5 checks; extended integer range; D_5 search
`scan_slices.py`, `summarize_scan.py`	Table 1
`integer_seed_search.py`	the bounded integer search of §4
`seeds_and_belyi.py`	Appendix A and the negative control

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- The PARI Group, *PARI/GP*; FLINT / python-flint; SymPy.

— End of note —